

SORA-SORT: A Simple Occlusion Risk-Aware Framework for Multi-Object Tracking

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Abstract

To cost-effectively acquire people flow data, the use of bird's-eye view cameras that capture wide areas from a distance is being considered. However, videos captured by such cameras frequently exhibit dynamic occlusions, such as those caused by interactions among individuals, and static occlusions, such as those caused by fixed obstacles, leading to a decline in tracking performance. Therefore, this study develops a system to estimate dynamic and static occlusion risk using only detection and tracking information, proposing a general multi-object tracking (MOT) performance improvement method by dynamically controlling tracker parameters based on this risk. Comparative experiments against a baseline method demonstrated that our approach outperformed the baseline on most metrics, suggesting that our method contributes to enhancing tracking performance in occlusion-prone areas. However, an increase in ID switches (IDS) and a decrease in processing speed occurred. The increase in IDS is considered to be due to the unnecessary retention of tracks that should have been deleted due to lifespan extension. This result implies a trade-off between tracking continuity and the consistency of ID assignment.

CCS Concepts

• Computing methodologies → Tracking.

Keywords

Multi-object tracking, people flow analysis, bird's eye view camera, occlusion

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1 Introduction

Multi-Object Tracking (MOT) is a task of detecting and tracking multiple objects in a video. People flow analysis is an important task for disaster prevention and smart city development. When using cameras for people flow tracking, the video should be processed on edge devices to protect privacy. However, edge devices have limited computational resources, so both tracking performance and computational efficiency are required.

Bird's-eye view cameras, which can capture images as shown in Figure 1, offer cost-effective wide-area coverage for people flow analysis but suffer from frequent occlusions, degrading tracking performance. Therefore, in conventional tracking methods, a decline in tracking performance often occurs due to primarily two reasons in data association. First is the accumulation of position estimation errors. In crowded scenes, long-term occlusions due to overlapping individuals occur frequently, making the predicted position error more likely to accumulate. The second is inappropriate lifespan management. Most tracking methods delete tracks that have not been tracked for a certain period. If a target cannot be detected due to a temporary occlusion, and the lifespan is exceeded before the target reappears, the track will be prematurely deleted. Consequently, a new ID will be assigned upon redetection (fragmentation). Due to these factors, while people flow analysis using bird's-eye view cameras faces the degrading lower tracking performance compared to analysis using general street cameras.

Based on the above, we aim to improve real-time people flow tracking performance on edge devices in environments with frequent occlusions. One potential approach to address the problem of decreased tracking performance in crowded videos from bird's-eye view cameras is to identify the occlusion risk for each area and

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Figure 1: Example image from a bird's-eye view camera.

modify the processing accordingly. One existing method for detecting occlusion risk is using image segmentation [9] [21]. However, it requires training on large datasets with extensive examples for each diverse class, which can lead to generalizability issues. Furthermore, its use of neural networks significantly increases computational load, making it unsuitable for real-time processing on edge devices.

This research proposes an approach that estimates areas with a high risk of occlusion online using only detection and tracking information. Based on this estimation, the period until a track that could not be tracked is deleted (lifespan) is dynamically adjusted. Since this method uses only detection and tracking information, it offers superior generalizability and processing speed compared to existing methods such as semantic segmentation. The research aims to integrate this estimation of high occlusion risk and the adaptive parameter control module into existing tracking methods and to validate its effectiveness from the perspectives of tracking performance and processing speed.

2 Related works

Suppressing tracking performance degradation caused by occlusions and intersections is a crucial research theme in MOT.

The work by Bewley et al. [2] proposed a lightweight and online MOT method (SORT). However, since the Kalman filter employed for position estimation is based on linear prediction, a decline in position estimation performance during abrupt changes in direction is an issue. Furthermore, failures in re-identification (Re-ID) when occlusions by people or static obstacles occur, as well as ID switches at intersections, are also significant difficulties.

As attempts to complement for objects missed by the detector, methods include applying single-object tracking (SOT) algorithms to individual targets [7] [6] or estimating the center of the overlapping positions between objects to complement detections [11]. Furthermore, methods introducing semantic segmentation into MOT are proposed as an approach to leverage more detailed object shape information.

Some methods utilize confidence scores of detection results [8] [23]. These methods aim to enhance robustness under noisy detections by using confidence scores to weight position estimation [8] prioritize data association [8] [23], or as criteria for updating appearance models [13].

In addition to motion information, there are methods to enhance the consistency of ID assignment by incorporating appearance information into the data association [19] [13] [11]. Adaptive methods for adjusting the weights of appearance and motion information in data association, based on the discriminativeness of appearance features, have also been proposed [13]. To address the high annotation costs for training conventional supervised Re-ID models, studies have emerged that propose unsupervised Re-ID learning [11]. Furthermore, a dynamic update to the appearance model based on observation quality [13] was also proposed.

Methods to capture object dynamics and contextual information between trajectories are also proposed. These include studies employing RNN or LSTM to model object motion patterns and temporal dependencies between trajectories [22] [18]. Furthermore, attention mechanisms that enable adaptive focus on salient features crucial for tracking have been exploited [5] [7]. Moreover, some methods utilize spatial-temporal relationships among objects [20] [12] [7].

As outlined above, MOT research has consistently addressed the challenge of mitigating tracking performance degradation due to occlusions and intersections. While methods utilizing appearance features and neural networks can enhance tracking performance, they often incur significant computational costs. Therefore, lightweight approaches such as methods based on SORT remain vital for real-time processing on edge devices.

Existing MOT methods do not control tracker parameters considering the scene's occlusion characteristics, which negatively affects tracking performance. For instance, if a track becomes unobservable due to a temporary occlusion, it may be prematurely deleted upon exceeding its fixed lifespan. This frequently leads to fragmentation and a degradation in tracking performance, particularly in areas prone to occlusion. Addressing these challenges necessitates the detection of occlusions. However, approaches such as estimating inter-object overlap using CNN [11] or employing semantic segmentation for detection [17] [3] significantly increase time complexity. Furthermore, image segmentation methods typically require extensive pre-training on large, diverse datasets, entailing high dataset creation costs and potentially limiting generalizability.

3 Methodology

The method integrates new modules into the framework of existing SORT-based trackers. Our method estimates the occlusion characteristics online using only detection and tracking information, and adaptively adjusts the tracking parameters based on this estimation. The main novelties are estimating scene-specific occlusion risk faster and with higher generalizability, and dynamically adjusting lifespan based on region occlusion risk. This improves the robustness of tracking especially in severe occlusion environments without significantly degrading execution speed. Figure 2 shows an overview of the system of this research, integrated into a general SORT-based model.

Our method consists of three main components: a dynamic occlusion heatmap, a static occlusion heatmap, and an adaptive parameter control module. These correspond to "Dynamic Occlusion Heatmap", "Static Occlusion Heatmap", and "State Management"

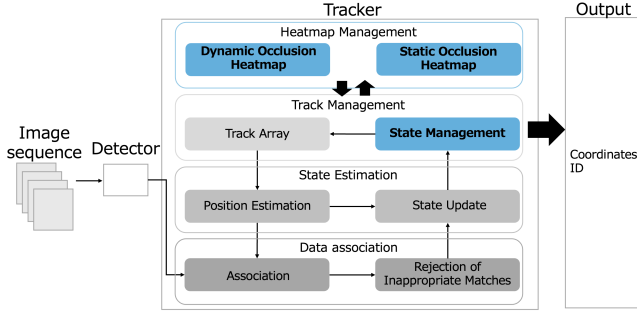


Figure 2: System architecture. Our modules' affected parts are shown with blue background and bold letters.

in the diagram, respectively. The dynamic occlusion heatmap estimates the risk of dynamic occlusions, such as pedestrians crossing paths. The static occlusion heatmap estimates the risk of static occlusion, such as occlusion by buildings. The state management module control track's lifespan dynamically based on occlusion risks estimated by the two modules. These modules prevent inappropriate deletion of tracks due to fixed lifespan during occlusion and improve tracking performance.

3.1 Dynamic Occlusion Heatmap

To estimate dynamic occlusion risk, the entire image is divided into tiles, and two heatmaps are managed: M_{det} , recording the number of detections per region, and M_{occ} , recording tracking failure events(deleted or newly appeared tracks) per region. These are events that occur frequently when tracking fails. The values in these heatmaps decay with time to prioritize recent information. The occlusion occurrence tendency $O(t_x, t_y)$ in an image region (t_x, t_y) is defined as in Equation (1).

$$O(t_x, t_y) = \frac{N_{occ}(t_x, t_y)}{N_{det}(t_x, t_y) + \epsilon} \quad (1)$$

$N_{occ}(t_x, t_y)$ is the value of M_{occ} in the image region (t_x, t_y) , $N_{det}(t_x, t_y)$ is the value of M_{det} in the same image region, and ϵ is a small positive constant to prevent division by zero. A tile with a high $O(t_x, t_y)$ value indicates a high sum of tracking failure events relative to the number of detections, and is thus inferred to be a region with a high occlusion risk. This occlusion occurrence tendency $O(t_x, t_y)$, associated with the coordinates of the divided regions, is defined as the dynamic occlusion heatmap. Figure 3 shows an example of a dynamic occlusion heatmap.

3.2 Static Occlusion Heatmap

To estimate static occlusions, a static occlusion heatmap is generated at regular intervals. The generation procedure is divided into three stages. First, $O(t_x, t_y)$ is binarized. This distinguishes static occlusions, characterized by extremely high tracking failure rates relative to detections, from dynamic occlusions in crowded areas. Next, for the binarized map, a closing operation, which involves dilation followed by erosion, is performed to shape the regions within the static obstacle areas. Then, a dilated map and an eroded map are created from this map, and a morphological gradient is calculated by taking the difference between these two maps. The outlines of



Figure 3: An example of dynamic occlusion heatmap. Tiles closer to deep purple indicate lower dynamic occlusion risk, while tiles closer to bright yellow indicate higher dynamic occlusion risk.

the static obstacles are inferred from this gradient. Figure 4 shows an example of a static occlusion heatmap.



Figure 4: An example of static occlusion heatmap. The boundaries of static obstacles are estimated and shown in red overlay.

3.3 Adaptive Parameter Control

When a track stops matching, its lifespan, starting from a base lifespan Δ_{base} , is dynamically extended based on its last observed position. This extension period consists of an extension period $\Delta_{dynamic}$ based on the occlusion occurrence tendency $O(t_x, t_y)$ and an extension period Δ_{static} based on the static occlusion heatmap. Specifically, first, the occlusion occurrence tendency $O(x_{last}, y_{last})$ at the last observed position is obtained from the dynamic occlusion heatmap, and the extension period $\Delta_{dynamic}$ based on the occlusion tendency is calculated using Equation (2). $\lambda_{dynamic}$ represents the factor for calculating the extension period based on the dynamic occlusion heatmap.

$$\Delta_{dynamic} = \lfloor O(x_{last}, y_{last}) \times \lambda_{dynamic} \rfloor \quad (2)$$

Next, the extension period Δ_{static} , based on whether the track is in the vicinity of an obstacle statically, is calculated using Equation

(3). λ_{static} represents the factor for calculating the extension period based on the static occlusion heatmap.

$$\Delta_{\text{static}} = \begin{cases} \lfloor O(x_{\text{last}}, y_{\text{last}}) \times \lambda_{\text{static}} \rfloor, & \text{if } \mathbf{B}_{\text{mask}}(t_{x_{\text{last}}}, t_{y_{\text{last}}}) = \text{True} \\ 0, & \text{else} \end{cases} \quad (3)$$

Finally, the final extension period is calculated using Equation (4). Δ_{total} is reset to the original lifespan Δ_{base} when the track matches again.

$$\Delta_{\text{total}} = \Delta_{\text{base}} + \Delta_{\text{dynamic}} + \Delta_{\text{static}} \quad (4)$$

4 Experimental evaluation

4.1 Datasets and Implementation

To validate the effectiveness of our method, we conducted comparative experiments between the baseline method OC-SORT [4] and a tracker that implements our method in addition to OC-SORT [4]. The Head Tracking 21 public dataset [14] [16], which features crowded scenes similar to those from bird’s-eye view cameras and frequent occlusions, was used as the dataset. We compared only trackers using the same detection results, called public detection. A Core™ i7-11800H CPU and 16 GB of memory were used. The tile size for both heatmaps was set to 120.

4.2 Metrics

As evaluation metrics, we used major metrics, Multi-Object Tracking Precision (MOTP) and Multi-Object Tracking Accuracy (MOTA) [1], False Negative ID (IDFN), Identification Precision (IDP), Identification Recall (IDR), and IDF1 [15], and ID Switches (IDS) proposed [10], to evaluate different aspects of the detection and tracking performance. We also evaluated processing speed shown as Hz.

4.3 Results and Discussion

Table 1 shows the evaluation results of the baseline and proposed methods on the 4 sequences of the HT21 dataset. As a result, our method outperformed the baseline in most metrics except for IDS and Hz. Mean increases were observed for MOTA by 5.8 points, MOTP by 30.9 points, IDF1 by 3.1 points, IDP by 1.9 points, IDR by 5.1 points, IDS by 881.4%. In contrast, the mean for IDFN decreased by approximately 18.5%. The processing speed of our method when processing all sequences decreased by approximately 41.0%. Our method showed significant improvements compared to the baseline, especially in IDFN and MOTP. On the other hand, it was inferior to the baseline in terms of IDS and Hz.

Table 1: Quantitative Evaluation Results

Dataset	Model	MOTA↑	MOTP↑	IDF1↑	IDP↑	IDR↑	IDS↓	IDFN↓	Hz↑
HT21-01	OC-SORT	69.4	34.9	79.2	96.4	67.1	55	6509	—
HT21-01	ours	73.8	59.7	81.6	99.6	72.4	752	4862	—
HT21-02	OC-SORT	57.3	45.7	66.6	91.0	52.5	2721	310231	—
HT21-02	ours	62.2	81.2	67.0	90.6	55.7	28704	248274	—
HT21-03	OC-SORT	30.4	40.3	40.9	86.6	26.8	1493	178141	—
HT21-03	ours	37.8	71.5	47.1	90.3	33.6	13175	147343	—
HT21-04	OC-SORT	36.2	40.9	49.6	92.5	33.9	861	111134	—
HT21-04	ours	42.5	73.0	52.9	93.4	38.9	7736	93204	—
mean	OC-SORT	48.3	40.5	59.1	91.6	45.1	1283	151504	39.7
mean	ours	54.1	71.4	62.2	93.5	50.2	12592	123421	23.4

The improvements in IDFN, a measure quantifying missed detections or misidentifications of ground truth objects, and MOTP, a measure of the average localization precision, are discussed below.

The improvement in IDFN is considered to be due to the effect of extended lifespan based on the two heat maps. By extending the lifespan of tracks that are likely to be under occlusion, tracks that would have been prematurely deleted in the conventional method survive. This increased the opportunity to continue tracking when the target reappears, thereby suppressing track fragmentation.

The improvement in MOTP is attributed to the prevention of fragmentation from temporary occlusions. This increases the amount of past trajectory information available for position estimation, which is considered to have improved position estimation performance.

The most significant problem in our method is the increase in IDS. This is attributed to an increase in overly persistent tracks (referred to as zombie tracks in this paper) due to lifespan extension. Particularly in crowded areas, it is considered that the IoU between the estimated position of a zombie track and another object co-incidentally becomes high, leading to incorrect association. This suggests that a trade-off exists between tracking continuity, which is related to lifespan, and the consistency of ID assignment.

5 Conclusions and Future Work

This research addressed tracking performance degradation due to occlusions by proposing a novel method that estimates occlusion risk online and adaptively adjusts the lifespan based on the occlusion risk. Our approach outperformed the baseline on most evaluation metrics, with particularly notable improvements in MOTP and significant reductions in IDFN, indicating enhanced tracking performance in occlusion-prone areas. However, an increase in IDS and decreased processing speed were observed, highlighting a trade-off between tracking continuity and the consistency of ID assignment, and the need to reduce computational cost for real-time applications.

Regarding the increase in ID switches, it is conceivable to investigate whether applying tracking methods that use appearance features can suppress the increase in ID switches while achieving improved tracking performance. Furthermore, instead of simple combination, further improvements could potentially be achieved by dynamically adjusting the weights of motion and appearance association costs according to the occlusion risk estimated by our method, or by adaptively controlling the Re-ID search range and thresholds when objects appear from occlusion-prone areas.

Additionally, to validate the effectiveness of our method in more detail, experiments using datasets with more static occlusions than the Head Tracking 21 dataset [14] [16] used in this research could be conducted.

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